

03. Integration Matrix v4.91 FULL CORRECTED

Полная матрица интеграций, событий и gate rules

1. Main Integration Flows

ID	Source	Target	Direction	Data Objects	Trigger	Protocol / Interface	Owner	Validation / Acceptance
F01	CRM / 1C	imos PLAN / imos iX	blue	OrderID, CustomerID, ProjectID, due date, payment status	Commercial order approved	API / file / 1C exchange	1C + imos	Order visible in engineering
F02	imos iX	iX Integration Server	blue	ProductID, ModuleID, PartID, BOM, BOO, ST, CAM, EngineeringVersionID	Engineering ready	imos Integration Server	imos integrator	EDP validation passed
F03	iX Integration Server	Engineering Data Package	blue	Normalized payload	Publish event	API/file/DB	imos integrator	Package versioned and traceable
F04	Engineering Data Package	MES One	blue	Products, parts, routing, operations, label base, hardware BOM	Production release	API/file/DB	MES One + imos	Work orders created
F05	Engineering Data Package	APS-lite / BMLT / CTP	blue	BOM, BOO, ST, routing, constraints	Simulation request	API/DB	MES/APS	BMLT/CTP calculated
F06	Engineering Data Package	Cut Rite V11.04	blue	Part list, materials, sizes, constraints	Optimization request	CSV/API/file	MES + Cut Rite	Optimization inputs accepted
F07	MES One	Cut Rite V11.04	blue	Approved Cutting Pool / Optimization Request	Release cutting pool	API/CSV/file	MES One	Optimization started
F08	Cut Rite V11.04	MES One	blue	CuttingBatchID, CutMapID, SheetID, PartID Placement, OffcutID, Not Nested Parts, Label Placement Map	Optimization completed	API/CSV/XML/file	Cut Rite + MES	All PartIDs resolved
F09	MES One / Cut Rite	HFE 200/22	blue	Label Placement Map, LabelID, coordinates, rotation, feed direction	Sheet picked to labeling	HOMAG interface / file / API	HOMAG + MES	HFE receives data
F10	HFE 200/22	MES One / UNS	red dashed	LabelApplied, LabelApplicationFailed	Labeling completed/failed	Collector / API / event	HOMAG/MES	Event stored and linked to SheetID
F11	woodStore / STORETEQ S-200	MES One / UNS	red dashed	SheetPicked, SheetDeliveredToLabeling, SheetDeliveredToNesting, Stock/Offcut facts	Sheet movement	woodStore interface / collector	HOMAG/MES	Sheet movement visible
F12	N-700 / HOMAG	MES One / UNS	red dashed	PartSeparated, OperationFactID, MachineID, ProgramID, Status	Cutting completed	MMR/OPC UA/collector	HOMAG/MES	Part becomes WIP object
F13	S-500 / BHX / V-200	MES One / UNS	red dashed	OperationStarted/Completed, Alarms, Downtime, PartID	Processing events	MMR/OPC UA/collector	HOMAG/MES	OperationFactID created
F14	Morbidelli X-400	Maestro Connect	blue/red	Telemetry, OEE support	Machine telemetry	Maestro Connect	SCM	OEE data available
F15	Morbidelli X-400	MES One Collectors -> UNS -> MES One	red dashed	OperationFactID, PartID, ProgramID, Status, Alarms	Operation execution	Local interface / collector	SCM/MES	Business facts independent of Maestro
F16	MES One	WMS-lite / Hardware	blue	KitRequired, Hardware BOM, ModuleID, due date	Parts ready or planned assembly	API/DB	MES/WMS	Picking task created
F17	WMS-lite / Hardware	MES One / UNS	red dashed	KitPrepared, KitChecked, KitIssuedToAssembler, KitShortage	Picking/check/issue	Scanner/API/event	WMS/MES	Assembly gate evaluated
F18	Assembly	MES One / UNS	red dashed	AssemblyStarted, AssemblyCompleted, AssemblyQCCompleted	Assembly execution	MES terminal/scanner	MES	Assembly facts visible
F19	Packing	MES One / WMS-lite / UNS	red dashed	Packed, PackageID, ShippingLabelApplied, PackingListPrinted	Packing completed	Scanner/printer/API	WMS/MES	Shipment gate evaluated
F20	Shipment / Delivery / Installation	MES One / UNS / BI	red dashed	ShipmentID, DeliveryID, InstallationTaskID, WorkAcceptanceActID	Shipment to closure	Mobile app/API	Logistics/Field Service	Order closed with act
F21	UNS / Event Store	BI / AI / APS-lite	orange	Facts, WIP, OEE, KPI, delays, risks	Event stream	MQTT/Sparkplug B / DB	IT/BI	Dashboard and recommendations

2. UNS Events

Event	Source	Target	Key Fields	Acceptance
LabelPlacementMapCreated	Cut Rite/MES One	HFE	SheetID, CutMapID, PartID, X/Y, rotation	HFE can print labels
LabelApplied	HFE	MES One/UNS	LabelID, PartID, SheetID	Traceability updated
LabelApplicationFailed	HFE	MES One/UNS	LabelID, error	N-700 release blocked
PartSeparated	N-700	MES One/UNS	PartID, OperationFactID	WIP object created
OffcutCreated	Cut Rite/woodStore	MES/WMS	OffcutID, dimensions	Return/sale/disposal flow
NotNestedPartsReturned	Cut Rite	MES One	PartID list, reason	Replan triggered
KitPrepared	WMS	MES/UNS	HardwareKitID	Gate input true
KitChecked	WMS	MES/UNS	HardwareKitID, checker	Gate input true
KitIssuedToAssembler	WMS	MES/UNS	HardwareKitID, AssemblerID	Gate input true
Packed	Packing	MES/UNS	PackageID	Shipment gate input
ShippingLabelApplied	Packing	MES/UNS	ShippingLabelID	Shipment gate input
PackingListPrinted	Packing	MES/UNS	PackingListID	Shipment gate input

3. Gate Rules

Gate	Required Conditions	If failed	Acceptance
Production Release	Engineering Ready + Material Ready + CAM Ready	Block production	Production order cannot be released
HFE to N-700	LabelApplied for required labels	Block nesting/cutting	No unlabeled board is cut
Assembly Permission Gate	PartsReadyForAssembly + KitPrepared + KitChecked + KitIssuedToAssembler	Block AssemblyStarted	AssemblyStarted rejected
Shipment Permission Gate	AssemblyCompleted + Packed + ShippingLabelApplied + PackingListPrinted	Block ReadyForShipment	ReadyForShipment rejected